Mock STEP Paper

March 22, 2026

- I think these questions are of a similar difficulty and style to STEP. I could also be wrong.
- There is no mechanics. I don't do the mechanics questions, so I am not capable of writing them unfortunately.
- Note that these likely contain more of the question types where the question builds up to some interesting result than would be realistic. These questions are more fun to write!
- Some questions are adapted from videos or other exams.
- A mark scheme is at the end. Hopefully it is correct.

Question 1

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ for integers $n \geqslant 0$.

(i) Derive a formula for I_n in terms of I_{n-2} valid for $n \geqslant 2$.

(ii) Hence, deduce expressions for I_{2k} and I_{2k+1} in terms of k, where k is a positive integer. You do not need to simplify your results.

(iii) Explain why $I_{2k+1} < I_{2k} < I_{2k-1}$ for $k \geqslant 1$.

Hence, deduce that:

$$1 < \frac{I_{2k}}{I_{2k+1}} < \frac{2k+1}{2k}$$

(iv) Use the results above to find $\lim_{k\to\infty} \frac{I_{2k}}{I_{2k+1}}$, and deduce that:

$$\frac{\pi}{2} = \lim_{k\to\infty} \left(\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdots \frac{2k}{2k-1} \cdot \frac{2k}{2k+1} \right)$$

Question 2

Define the following:

- $T(x) = \dfrac{ax+b}{cx+d}$, where a, b, c, and d are real constants.
- $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.
- For a sequence, u_n the recurrence relation $u_{n+1} = T(u_n)$ holds for integers $n \geqslant 0$, with a given initial value u_0.
- For sequences x_n and y_n, the matrix equation:

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} = M^n \begin{pmatrix} u_0 \\ 1 \end{pmatrix}$$

  holds.

(i) Prove by induction that $u_n = \frac{x_n}{y_n}$ for all integers $n \geqslant 0$.

(ii) Suppose that M can be written in the form $M = \lambda I + N$, where λ is a scalar constant, I is the 2×2 identity matrix, and N is a matrix such that $N^2 = \mathbf{0}$ (the zero matrix).

Prove that for all integers $n \geqslant 1$:

$$M^n = \lambda^n I + n\lambda^{n-1} N$$

Carefully justify your reasoning.

(iii) Consider the specific sequence defined by $u_{n+1} = \dfrac{7u_n - 9}{u_n + 1}$.

Write down the associated matrix M, and find the value of λ and the matrix N such that $M = \lambda I + N$ and $N^2 = \mathbf{0}$.

(iv) Show that, for the sequence defined in (iii):

$$u_n = \frac{4u_0 + 3n(u_0 - 3)}{4 + n(u_0 - 3)}$$

(v) Show that if $u_0 = 2$, the sequence is undefined at $n = 4$.

(vi) State the condition on u_0 for which the sequence u_n is well-defined for all integers $n \geqslant 0$.

Question 3

The notation $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x, and $\lceil x \rceil$ denotes the smallest integer greater than or equal to x.

(i) By considering the cases where n is even and n is odd, or otherwise, show that:

- $\lceil n/2 \rceil - 1 = \lfloor (n-1)/2 \rfloor$ for all integers $n \geqslant 1$.
- $n - \lfloor n/2 \rfloor = \lceil n/2 \rceil$ for all integers $n \geqslant 0$.

(ii) A positive integer n is said to be *palindromic* if its base-10 representation reads the same forwards and backwards. For example, the number 12321 would be palindromic.

Let P_d denote the set of all palindromic numbers with exactly d digits, and let $n(d)$ be the number of elements in P_d.

Use a combinatorial argument to show that $n(d) = 9 \times 10^{\lfloor (d-1)/2 \rfloor}$ for all integers $d \geqslant 1$.

(iii) Let S_d be the sum of the reciprocals of the numbers in P_d.

Show that $S_d < 9 \times 10^{-\lceil (d-1)/2 \rceil}$ for all integers $d \geqslant 2$.

(iv) Let S be the sum of the reciprocals of all palindromic numbers:

$$S = \sum_{d=1}^{\infty} S_d$$

Show that the series for S converges and that $S < 11$. You should consider the sum $S_1 = \sum_{k=1}^{9} \frac{1}{k}$ separately using a similar argument.

(v) By considering the leading digit $m \in \{1, 2, \ldots, 9\}$ of the numbers in P_d to obtain an expression for S_n in terms of S_1, and given that $S_1 < \frac{11}{4}$, show that:

$$S < \frac{121}{36}$$

Question 4

Consider the differential equation:

$$\frac{dy}{dx} + R(x)y^2 + P(x)y + Q(x) = 0$$

where $R(x)$, $P(x)$, and $Q(x)$ are given functions and $R(x) \neq 0$.

(i) Show that the substitution:

$$y = \frac{1}{R(x)u}\frac{du}{dx}$$

transforms the equation into the linear second-order differential equation:

$$\frac{d^2u}{dx^2} + \left(P(x) - \frac{R'(x)}{R(x)}\right)\frac{du}{dx} + Q(x)R(x)u = 0$$

(ii) By using the substitution $u = x^k$, or otherwise, show that the general solution to the differential equation:

$$\frac{dy}{dx} + xy^2 - \frac{2}{x}y - \frac{5}{x^3} = 0, \quad x > 0$$

is $y = \dfrac{Ax^6 - B}{x^2(Ax^6 + B)}$, where A and B are arbitrary constants.

(iii) Find the particular solution, given that $y(0) = 1$, to the following differential equation:

$$\frac{dy}{dx} + e^x y^2 - y - 2e^{-x} = 0$$

Express your final answer in the form $y = e^{-x}\left(1 + k\tanh(kx)\right)$, where k is a constant to be determined.

Question 5

A family of lines L_t is defined by the equation:

$$y = tx - \frac{1}{3}t^3$$

where t is a real parameter.

(i) By considering the turning points of the function $f(t) = \frac{1}{3}t^3 - Xt + Y$, show that exactly two distinct lines in L_t pass through a point (X, Y) if and only if the point lies on the curve C given by the Cartesian equation:

$$Y^2 = \frac{4}{9}X^3$$

(ii) Calculate the exact area of the region bounded by the curve C and the vertical line $X = 3$.

(iii) A second family of straight lines L_m is defined by the equation:

$$y = mx - \frac{1}{4}m^4$$

where m is a real parameter.

Show that the points (X, Y) through which exactly two lines in L_m pass lie on the curve D given by:

$$Y^3 = \frac{27}{64}X^4$$

(iv) Find the exact coordinates (X, Y) of the point of intersection, other than the origin, of the curve D and the branch of the curve C where $Y > 0$, giving your answers in the form $\frac{3^a}{2^b}$ where a and b are integers.

Hence, find the exact value of $\tan\theta$, where θ is the acute angle between the tangents to C and C' at the point P. Give your answer as a fraction in its simplest form.

Question 6

(i) Show that the area bounded by the curve $y = \cosh x$, the x-axis, the y-axis, and the line $x = a$ (where $a > 0$) is exactly equal to the arc length of the curve from $x = 0$ to $x = a$.

Conversely, suppose $f(x)$ is a function such that $f(x) > 0$ and $f'(x) > 0$ for all $x \geqslant 0$. Given that the area under the curve $y = f(x)$ from $x = 0$ to $x = t$ is equal to the arc length over the same interval for all $t > 0$, find all such functions.

(ii) A uniform chain hangs in the shape of a generalised catenary given by the equation $y = c\cosh\left(\frac{x}{c}\right)$, where c is a positive constant.

Let A_c be the area under the curve from $x = 0$ to $x = a$, and let L_c be the arc length over the same interval. Find A_c and L_c, and hence state the relationship between them in terms of c.

(iii) Let $P(x, y)$ be a point on the curve $y = \cosh x$. The normal to the curve at P intersects the x-axis at the point G.

Find the coordinates of G, and hence show that the length of the line segment PG is equal to y^2.

(iv) The radius of curvature, ρ, of a curve is given by the formula:

$$\rho = \frac{(1 + (y')^2)^{3/2}}{|y''|}$$

Show that for the curve $y = \cosh x$, the radius of curvature at P is exactly equal to the length PG.

Question 7

The generalised triangle inequality for complex numbers is given by:

$$|z_1 + z_2 + \cdots + z_n| \leqslant |z_1| + |z_2| + \cdots + |z_n|$$

(i) Consider the polynomial $P(z) = 3z^2 + 2z + 1$. By considering the expression $(1 - z)P(z)$, prove by contradiction that all roots of $P(z)$ satisfy $|z| \leqslant 1$.

(ii) Let $f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$, where the coefficients are real numbers satisfying:

$$a_n > a_{n-1} > \cdots > a_1 > a_0 > 0$$

By considering $(1 - z)f(z)$, prove that all roots of $f(z)$ must lie within or on the unit circle in the complex plane.

(iii) Let $g(z) = a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n$, where the coefficients satisfy the same inequalities $a_n > a_{n-1} > \cdots > a_0 > 0$.

By using a suitable substitution, deduce that all roots of $g(z)$ must satisfy $|z| \geqslant 1$.

(iv) Consider the polynomial $R(z) = z^4 + 3z^3 + 8z^2 + 24z + 32$.

By using a substitution of the form $z = \lambda w$ for some positive integer λ, find the smallest integer value of λ such that the theorem proven in part (ii) can be directly applied.

Hence, state an upper bound for the modulus of the roots of $R(z)$, and find a lower bound for the modulus of the roots of the polynomial $S(z) = 32z^4 + 24z^3 + 8z^2 + 3z + 1$.

Question 8

Let $a_0 = \sqrt{2} + \sqrt{3} + \sqrt{6}$, and let $a_{n+1} = \dfrac{a_n^2 - 5}{2(a_n + 2)}$ for integers $n \geqslant 0$.

(i) Prove by induction that

$$a_n = \cot\left(\frac{2^{n-3}\pi}{3}\right) - 2$$

for all $n \geqslant 0$.

(ii) Determine whether the sequence a_n converges. Carefully justify your answer.

Question 9

The random variable X has probability density function

$$f(x) = \begin{cases} Cx^2(1-x)^n & 0 \leqslant x \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

where n is an integer greater than 2.

(i) Find the mean $\mu = E(X)$ and the mode M of the distribution in terms of n only.

(ii) Show that $\mu > M$ for all $n > 2$.

(iii) By considering an area bounded by $x = \mu$, show that the median m satisfies $\mu < m < M$ for $n = 3$.

(iv) Explain why μ, M, and m will all approach zero for very large n.

Question 10

In this question, $a \geqslant 2$.

A bag contains n balls, of which a are red and b are blue.

Balls are drawn from the bag one at a time, at random and without replacement. Let X_1 be the number of draws required to obtain the first red ball, and X_r be the number of draws required to obtain the r-th red ball.

(i) Show that the probability $P(X_1 > k)$ is given by

$$P(X_1 > k) = \frac{\binom{n-k}{a}}{\binom{n}{a}}$$

for $0 \leqslant k \leqslant b$.

Using the identity $\sum_{j=a}^{n} \binom{j}{a} = \binom{n+1}{a+1}$, or otherwise, show that $E(X_1) = \frac{n+1}{a+1}$.

(ii) Using the identity $\sum_k P(X > k) = E(X)$, or otherwise, show that for $1 \leqslant r \leqslant a$:

$$E(X_r) = \frac{r(n+1)}{a+1}$$

(iii) Let $a = 2$. Find an exact expression for $P(X_2 - X_1 = k)$ in terms of n and k, where $1 \leqslant k \leqslant n-1$.

(iv) In a modified experiment, balls are drawn with replacement. Let Z_r be the number of draws required to obtain r red balls. State $E(Z_r)$ in terms of n and a. Show that as $n \to \infty$ and $a \to \infty$ such that the proportion $p = a/n$ remains constant, the difference $E(Z_r) - E(X_r)$ approaches 0.

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The mark scheme follows.

Mark Scheme

Question 1

Part (i) [4]

Writes $\sin^n x = \sin^{n-1} x \sin x$ and attempts IBP.

$I_n = \left[-\cos x \sin^{n-1} x\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\cos x (n-1) \sin^{n-2} x \cos x \, dx$. Boundary term evaluates to 0.

Substitutes $\cos^2 x = 1 - \sin^2 x$ into the integral to get: $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) \, dx$

Shows $I_n = (n-1) I_{n-2} - (n-1) I_n$, which simplifies to the required $I_n = \frac{n-1}{n} I_{n-2}$.

Part (ii) [6]

Repeated application of the reduction formula for I_{2k}.

$I_{2k} = \frac{2k-1}{2k} \cdot \frac{2k-3}{2k-2} \cdots \frac{1}{2} I_0$

Evaluates $I_0 = \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2}$ and states the final correct product for I_{2k}.

Repeated application of the reduction formula for I_{2k+1}.

$I_{2k+1} = \frac{2k}{2k+1} \cdot \frac{2k-2}{2k-1} \cdots \frac{2}{3} I_1$

Evaluates $I_1 = \int_0^{\frac{\pi}{2}} \sin x \, dx = 1$ and states the final correct product for I_{2k+1}.

Part (iii) [5]

For $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$.

Hence $\sin^{2k+1} x < \sin^{2k} x < \sin^{2k-1} x$.

Integrating preserves the inequality, so $I_{2k+1} < I_{2k} < I_{2k-1}$.

Division by I_{2k+1} (valid since $I_{2k+1} > 0$) to get $1 < \frac{I_{2k}}{I_{2k+1}} < \frac{I_{2k-1}}{I_{2k+1}}$.

Uses the reduction formula to substitute $I_{2k-1} = \frac{2k+1}{2k} I_{2k+1}$

Part (iv) [5]

Applies limit as $k \to \infty$ to the inequality from (iii): $1 \leqslant \lim_{k\to\infty} \frac{I_{2k}}{I_{2k+1}} \leqslant \lim_{k\to\infty} \frac{2k+1}{2k}$.

Evaluates the upper bound limit to find $\lim_{k\to\infty} \frac{I_{2k}}{I_{2k+1}} = 1$.

Substitutes the fractional products from (ii) into the ratio $\frac{I_{2k}}{I_{2k+1}}$.

Correct algebra.

Isolates $\frac{\pi}{2}$ and applies limit to deduce the required result.

Question 2

Part (i) [4]

Base case $n = 0$: $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = M^0 \begin{pmatrix} u_0 \\ 1 \end{pmatrix} = I \begin{pmatrix} u_0 \\ 1 \end{pmatrix} = \begin{pmatrix} u_0 \\ 1 \end{pmatrix}$. So $\frac{x_0}{y_0} = \frac{u_0}{1} = u_0$, which is true.

Assume true for $n = k$, so $u_k = \frac{x_k}{y_k}$. Consider $n = k + 1$:

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = M \begin{pmatrix} x_k \\ y_k \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_k \\ y_k \end{pmatrix} = \begin{pmatrix} ax_k + by_k \\ cx_k + dy_k \end{pmatrix}.$$

Then $\frac{x_{k+1}}{y_{k+1}} = \frac{ax_k+by_k}{cx_k+dy_k}$.

Divides the numerator and denominator by y_k to get $\frac{a(x_k/y_k)+b}{c(x_k/y_k)+d}$.

Substitutes the inductive hypothesis $x_k/y_k = u_k$ to obtain $\frac{au_k+b}{cu_k+d} = T(u_k) = u_{k+1}$, and states conclusion to complete the proof by induction.

Part (ii) [4]

States $M^n = (\lambda I+N)^n$ and notes that because the identity matrix I commutes with all matrices, the binomial expansion can be applied.

Expands using the binomial theorem: $M^n = (\lambda I)^n + \binom{n}{1}(\lambda I)^{n-1}N + \binom{n}{2}(\lambda I)^{n-2}N^2 + \dots$

Recognises that since $N^2 = \mathbf{0}$, all subsequent terms containing $N^2, N^3, \dots$ evaluate to the zero matrix.

Simplifies the remaining first two terms to strictly yield $M^n = \lambda^n I + n\lambda^{n-1}N$.

If no justification for using the binomial theorem, max two marks.

Part (iii) [4]

States the associated matrix is $M = \begin{pmatrix} 7 & -9 \\ 1 & 1 \end{pmatrix}$.

Sets up $N = M - \lambda I = \begin{pmatrix} 7-\lambda & -9 \\ 1 & 1-\lambda \end{pmatrix}$.

Computes N^2 and sets the bottom-left entry (or another suitable entry) to 0: $1(7-\lambda)+1(1-\lambda) = 0 \implies 8 - 2\lambda = 0$.

Solves to find $\lambda = 4$ and substitutes back to explicitly state $N = \begin{pmatrix} 3 & -9 \\ 1 & -3 \end{pmatrix}$ (optionally verifying $N^2 = \mathbf{0}$).

Part (iv) [5]

Attempts to substitute $\lambda = 4$ and N into the formula from (ii):

$$M^n = 4^n \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + n4^{n-1} \begin{pmatrix} 3 & -9 \\ 1 & -3 \end{pmatrix} = \begin{pmatrix} 4^n + 3n4^{n-1} & -9n4^{n-1} \\ n4^{n-1} & 4^n - 3n4^{n-1} \end{pmatrix}.$$

Multiplies M^n by $\begin{pmatrix} u_0 \\ 1 \end{pmatrix}$ to find expressions for x_n and y_n.

Forms the fraction $u_n = \frac{x_n}{y_n} = \frac{(4^n+3n4^{n-1})u_0 - 9n4^{n-1}}{n4^{n-1}u_0 + 4^n - 3n4^{n-1}}$.

Cancels the common factor of 4^{n-1} to simplify the expression to $u_n = \frac{4u_0+3n(u_0-3)}{4+n(u_0-3)}$.

Part (v) [1]

Substitutes $u_0 = 2$ to get $u_n = \frac{8-3n}{4-n}$ and notes that the denominator is 0 when $n = 4$, making the sequence undefined.

Part (vi) [2]

States that for the sequence to be well-defined for all $n \geqslant 0$, the denominator must never equal zero: $4 + n(u_0 - 3) \neq 0$.

Rearranges to isolate u_0 and states the final condition: $u_0 \neq 3 - \frac{4}{n}$ for all strictly positive integers n.

Question 3

Part (i) [3]

Tests the first identity using even/odd cases. For $n = 2k$, $\lceil 2k/2 \rceil - 1 = k - 1$ and $\lfloor (2k-1)/2 \rfloor = k - 1$. For $n = 2k + 1$, $\lceil (2k + 1)/2 \rceil - 1 = k$ and $\lfloor 2k/2 \rfloor = k$.

Tests the second identity similarly. For $n = 2k$, $2k - \lfloor 2k/2 \rfloor = k = \lceil 2k/2 \rceil$. For $n = 2k + 1$, $(2k + 1) - \lfloor (2k + 1)/2 \rfloor = k + 1 = \lceil (2k + 1)/2 \rceil$.

Award $(n - 1)$ marks where n is the number of cases correctly reasoned.

Part (ii) [5]

Argues that a palindrome of d digits is determined by its first $\lceil d/2 \rceil$ digits.

Identifies the number of choices: 9 for the first digit (cannot be 0) and 10 for each of the remaining $\lceil d/2 \rceil - 1$ digits.

States $n(d) = 9 \times 10^{\lceil d/2 \rceil - 1}$.

Applies the first identity from (i) to substitute the exponent: $\lceil d/2 \rceil - 1 = \lfloor (d - 1)/2 \rfloor$.

Final expression $n(d) = 9 \times 10^{\lfloor (d-1)/2 \rfloor}$ clearly shown.

Part (iii) [2]

Observes that the smallest d-digit number is 10^{d-1}, so for all $k \in P_d$, $1/k \leqslant 10^{-(d-1)}$.

Establishes $S_d \leqslant n(d) \times 10^{-(d-1)} = 9 \times 10^{\lfloor (d-1)/2 \rfloor - (d-1)}$.

Applies the second identity from (i) with $n = d - 1$ to simplify the exponent to $-\lceil (d-1)/2 \rceil$.

Part (iv) [4]

For $d = 1$, $S_1 = \sum_{k=1}^{9} \frac{1}{k} < \sum_{k=1}^{9} 1 = 9$.

Bounds terms for $d \geqslant 2$: $S_2 < 9(10^{-1})$, $S_3 < 9(10^{-1})$, $S_4 < 9(10^{-2})$, $S_5 < 9(10^{-2}) \ldots$

Sums the geometric series: $S < S_1 + 2(0.9 + 0.09 + 0.009 + \ldots)$.

Evaluates the sum as $S < 9 + 1.8 \times \frac{10}{9} = 11$. States convergence due to the comparison with a convergent geometric series.

Part (v) [6]

Partitions P_d by the leading digit $m \in \{1, 2, \ldots, 9\}$. Notes that for a fixed m, there are $10^{\lfloor (d-1)/2 \rfloor}$ such palindromes. (This is equivalent to $n(d)/9$).

For a fixed m, each palindrome $k \geqslant m \times 10^{d-1}$. Summing the reciprocals for a fixed m gives $\sum \frac{1}{k} < \frac{n(d)/9}{m \times 10^{d-1}}$.

Summing over all m yields $S_d < \frac{n(d)}{9 \times 10^{d-1}} \sum_{m=1}^{9} \frac{1}{m} = \frac{n(d)}{9 \cdot 10^{d-1}} S_1$.

Uses results from (i) to simplify the coefficient: $\frac{n(d)}{9 \cdot 10^{d-1}} = 10^{-\lceil (d-1)/2 \rceil}$.

Constructs the total sum $S < S_1 + S_1(2 \times 10^{-1} + 2 \times 10^{-2} + \ldots) = S_1(1 + \frac{2}{9}) = \frac{11}{9} S_1$.

Substituting $S_1 < \frac{11}{4}$ gives $S < \frac{11}{9} \times \frac{11}{4} = \frac{121}{36}$.

Question 4

Part (i) [10]

Attempts to differentiate $y = \frac{u'}{Ru}$.

Correctly obtains $y' = \frac{u''}{Ru} - \frac{(u')^2}{Ru^2} - \frac{R'u'}{R^2 u}$.

Substitutes the expressions for y and y' into the original DE.

Correctly evaluates $R(x)y^2 = R(x)\left(\frac{u'}{Ru}\right)^2 = \frac{(u')^2}{Ru^2}$.

Cancels non-linear terms $\pm \frac{(u')^2}{Ru^2}$.

$\frac{u''}{Ru} - \frac{R'u'}{R^2 u} + P\frac{u'}{Ru} + Q = 0$.

Multiplies by Ru.

Identifies and groups the $\frac{du}{dx}$ terms.

Correct result.

Part (ii) [5]

Identifies $R = x, P = -2/x, Q = -5/x^3$ and forms the equation: $u'' - \frac{3}{x}u' - \frac{5}{x^2}u = 0$.

Substitutes $u = x^k$ to form $k^2 - 4k - 5 = 0$.

Solves to find roots $k = 5, -1$, giving general solution $u = Ax^5 + Bx^{-1}$.

Differentiates u and substitutes u and u' into the definition $y = \frac{u'}{xu}$.

Simplifies the resulting fraction into the form $y = \frac{5Ax^6 - B}{x^2(Ax^6 + B)}$.

Part (iii) [5]

Identifies $R = e^x, P = -1, Q = -2e^{-x}$ and forms $u'' - 2u' - 2u = 0$.

Solves the auxiliary equation $m^2 - 2m - 2 = 0$ to get $m = 1 \pm \sqrt{3}$.

Writes $u = e^x(Ae^{\sqrt{3}x} + Be^{-\sqrt{3}x})$ and substitutes into $y = \frac{u'}{e^x u}$.

Applies the boundary condition $y(0) = 1$ to deduce $A = B$ from the resulting algebraic expression.

Substitutes $A = B$ and simplifies yielding $y = e^{-x}(1 + \sqrt{3}\tanh(\sqrt{3}x))$.

Question 5

Part (i) [5]

Differentiates to find turning points: $f'(t) = t^2 - X = 0 \implies t = \pm\sqrt{X}$.

States that exactly two distinct roots occur if one turning point is a root: $f(\pm\sqrt{X}) = 0$.

Substitutes $t = \sqrt{X}$ to find $Y = \frac{2}{3}X^{3/2}$.

Substitutes $t = -\sqrt{X}$ to find $Y = -\frac{2}{3}X^{3/2}$.

Squares either result to obtain $Y^2 = \frac{4}{9}X^3$.

Part (ii) [3]

Sets up the area integral, noting symmetry: Area $= 2\int_0^3 \frac{2}{3}X^{3/2}\,dX$.

Integrates correctly to find $[\frac{8}{15}X^{5/2}]_0^3$.

Evaluates at the limit: $\frac{8}{15}(3^{5/2}) = \frac{8}{15}(9\sqrt{3}) = \frac{24\sqrt{3}}{5}$.

Part (iii) [4]

Sets up $g(m) = \frac{1}{4}m^4 - Xm + Y$ and differentiates: $g'(m) = m^3 - X = 0$.

Finds the single real turning point $m = X^{1/3}$. Sets $g(X^{1/3}) = 0$ for a repeated root.

Substitutes to find $Y = \frac{3}{4}X^{4/3}$.

Cubes the result to obtain $Y^3 = \frac{27}{64}X^4$.

Part (iv) [8]

Equates Y from C and D: $\frac{2}{3}X^{3/2} = \frac{3}{4}X^{4/3} \implies X^{1/6} = \frac{9}{8}$.

Solves for X: $X = (\frac{3^2}{2^3})^6 = \frac{3^{12}}{2^{18}}$.

Solves for Y: $Y = \frac{2}{3}(\frac{3^{12}}{2^{18}})^{3/2} = \frac{2}{3} \cdot \frac{3^{18}}{2^{27}} = \frac{3^{17}}{2^{26}}$.

Differentiates both curves: $m_C = \frac{dY}{dX} = X^{1/2}$ and $m_D = \frac{dY}{dX} = X^{1/3}$.

Finds gradients $m_C = X_P^{1/2} = \frac{3^6}{2^9}$ and $m_D = X_P^{1/3} = \frac{3^4}{2^6}$.

Applies $\tan\theta = \left|\frac{m_C - m_D}{1 + m_C m_D}\right|$.

Simplifies numerator: $\frac{3^6}{2^9} - \frac{3^4}{2^6} = \frac{3^6 - 2^3 \cdot 3^4}{2^9} = \frac{3^4(3^2-8)}{2^9} = \frac{3^4}{2^9}$ or $\frac{2^6 \cdot 3^4}{2^{15}}$or evaluates to $\frac{81}{512}$ or $\frac{5184}{32768}$.

Simplifies denominator: $1 + \frac{3^6 \cdot 3^4}{2^9 \cdot 2^6} = 1 + \frac{3^{10}}{2^{15}} = \frac{2^{15}+3^{10}}{2^{15}}$ or evaluates to $\frac{91817}{32768}$.

Combines to get $\tan\theta = \frac{3^4}{2^9} \cdot \frac{2^{15}}{2^{15}+3^{10}} = \frac{3^4 \cdot 2^6}{2^{15}+3^{10}}$.

$\frac{81 \cdot 64}{32768+59049} = \frac{5184}{91817}$ or any other valid method to deal with the terrible numbers.

Question 6

Part (i) [8]

$A = \int_0^a \cosh x \, dx = [\sinh x]_0^a = \sinh a$.

$L = \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx = \int_0^a \sqrt{1 + \sinh^2 x} \, dx$.

Uses $\cosh^2 x - \sinh^2 x = 1$ to simplify: $L = \int_0^a \cosh x \, dx = \sinh a$. Concludes $A = L$.

Sets up $\int_0^t f(x) \, dx = \int_0^t \sqrt{1 + (f'(x))^2} \, dx$. Do not award mark if indefinite integral used.

Differentiates with respect to t to get $f(t) = \sqrt{1 + (f'(t))^2}$.

Squares and rearranges: $(f'(t))^2 = f(t)^2 - 1$. Since $f'(x) > 0$, $\frac{df}{dx} = \sqrt{f^2 - 1}$.

Separates variables: $\int \frac{1}{\sqrt{f^2-1}} \, df = \int 1 \, dx \implies \operatorname{arcosh}(f) = x + k$.

States the final form: $f(x) = \cosh(x + k)$ for a constant k.

Part (ii) [5]

$A_c = \int_0^a c\cosh\left(\frac{x}{c}\right)\, dx = \left[c^2 \sinh\left(\frac{x}{c}\right)\right]_0^a = c^2 \sinh\left(\frac{a}{c}\right)$.

$\frac{dy}{dx} = \sinh\left(\frac{x}{c}\right)$.

$L_c = \int_0^a \sqrt{1+\sinh^2\left(\frac{x}{c}\right)}\, dx = \int_0^a \cosh\left(\frac{x}{c}\right)\, dx = \left[c\sinh\left(\frac{x}{c}\right)\right]_0^a = c\sinh\left(\frac{a}{c}\right)$.

Identifies the relationship: $A_c = cL_c$.

Part (iii) [4]

Gradient of tangent $m_t = \sinh x$, so gradient of normal $m_n = -\frac{1}{\sinh x}$.

Forms equation of the normal at (x, y): $Y - \cosh x = -\frac{1}{\sinh x}(X - x)$.

Finds G by setting $Y = 0$: $G = (x + \sinh x \cosh x, 0)$.

Correctly calculates the length $PG = y^2$. I don't want to write out all the algebra but lets say it's worth two marks.

Part (iv) [3]

Notes that for $y = \cosh x$, $y' = \sinh x$ and $y'' = \cosh x = y$.

$\rho = \frac{(1+\sinh^2 x)^{3/2}}{\cosh x} = \frac{(\cosh^2 x)^{3/2}}{\cosh x} = \frac{\cosh^3 x}{\cosh x} = \cosh^2 x$.

Concludes that $\rho = y^2 = PG$.

Question 7

Part (i) [6]

$(1-z)P(z) = (1-z)(3z^2+2z+1) = 3z^2+2z+1-3z^3-2z^2-z = 1+z+z^2-3z^3$.

If z is a root of $P(z)$, then $P(z) = 0$, which implies $(1-z)P(z) = 0$. Thus, $3z^3 = 1+z+z^2$.

Takes the modulus and applies triangle inequality: $|3z^3| \leqslant |1| + |z| + |z^2| = 1 + |z| + |z|^2$.

Assumes for contradiction that there exists a root such that $|z| > 1$.

Since $|z| > 1$, then $|z|^2 > |z|$ and $|z|^2 > 1$. This implies $1 + |z| + |z|^2 < |z|^2 + |z|^2 + |z|^2 = 3|z|^2$.

Combinining, $3|z|^3 < 3|z|^2 \implies |z| < 1$. This contradicts $|z| > 1$, hence $|z| \leqslant 1$ for all roots.

Part (ii) [6]

Essentially the same marks as (i).

Part (iii) [3]

Substitutes $z = \frac{1}{w}$.

Substituting into $g(z) = 0$ and multiplying by w^n: $w^n[a_0(1/w)^n + a_1(1/w)^{n-1} + \cdots + a_n] = a_0 + a_1 w + \cdots + a_n w^n = 0$. This is the polynomial $f(w)$ from part (ii).

By part (ii), roots satisfy $|w| \leqslant 1$. Since $|w| = 1/|z|$, then $1/|z| \leqslant 1 \implies |z| \geqslant 1$.

Part (iv) [5]

Substitutes $z = \lambda w$ into $R(z)$: $R(\lambda w) = \lambda^4 w^4 + 3\lambda^3 w^3 + 8\lambda^2 w^2 + 24\lambda w + 32$.

Identifies coefficients: $a_4 = \lambda^4, a_3 = 3\lambda^3, a_2 = 8\lambda^2, a_1 = 24\lambda, a_0 = 32$. Requires $\lambda^4 > 3\lambda^3 > 8\lambda^2 > 24\lambda > 32$.

Tests integer values: If $\lambda = 3$: $81 > 81$ (False). If $\lambda = 4$: $256 > 192 > 128 > 96 > 32$ (True). Smallest integer $\lambda = 4$.

States $|w| \leqslant 1$, so $|z| = |4w| = 4|w| \leqslant 4$. Upper bound for modulus of roots of $R(z)$ is 4.

Observes that $S(z) = z^4 R(1/z)$. By the logic in part (iii), the roots of $S(z)$ are the reciprocals of the roots of $R(z)$. Lower bound for $|z|$ in $S(z)$ is $1/4$.

Question 8

Part (i) [10]

Base case $n = 0$: $\cot(\pi/24) - 2 = (\sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6}) - 2 = \sqrt{2} + \sqrt{3} + \sqrt{6}$.

Let $b_n = a_n + 2$. Recurrence: $b_{n+1} - 2 = \frac{(b_n - 2)^2 - 5}{2b_n} \implies b_{n+1} = \frac{b_n^2 - 1}{2b_n}$.

$\cot(2\theta) = \frac{\cot^2\theta - 1}{2\cot\theta}$.

Assume $b_k = \cot(\theta_k)$ for $\theta_k = \frac{2^{k-3}\pi}{3}$. Then $b_{k+1} = \frac{\cot^2\theta_k - 1}{2\cot\theta_k} = \cot(2\theta_k)$.

$2\theta_k = 2 \cdot \frac{2^{k-3}\pi}{3} = \frac{2^{(k+1)-3}\pi}{3}$.

$a_n = b_n - 2 = \cot\left(\frac{2^{n-3}\pi}{3}\right) - 2$ for all $n \geqslant 0$.

Let's say all except the first and last step are worth two marks to get up to ten.

Part (ii) [3]

Sets $\cot\left(\frac{2^{n-3}\pi}{3}\right) = \sqrt{3}$.

$\frac{2^{n-3}\pi}{3} = \frac{\pi}{6} \implies 2^{n-3} = \frac{1}{2} \implies n - 3 = -1$.

Deduces $n = 2$.

Part (iii) [7]

Investigates $\theta_n = \frac{2^{n-3}\pi}{3}$ for $n \geqslant 3$. For $n = 3$, $\theta_3 = \pi/3 \implies a_3 = \frac{1}{\sqrt{3}} - 2$.

For $n = 4$, $\theta_4 = 2\pi/3 \implies a_4 = -\frac{1}{\sqrt{3}} - 2$.

$\theta_{n+2} = \theta_n + 2^{n-3}\pi$.

For $n \geqslant 3$, 2^{n-3} is an integer, so $a_{n+2} = a_n$. (two marks)

Sequence is $a_1, a_2, a_3, a_4, a_3, a_4, \ldots$.

Since $a_3 \neq a_4$, the sequence oscillates and does not converge.

Question 9

Part (i) [8]

States that $\displaystyle\int_0^1 f(x)\,dx = 1$.

$\displaystyle\int_0^1 x^2(1-x)^n\,dx = \frac{2}{(n+1)(n+2)(n+3)}$, so $C = \dfrac{(n+1)(n+2)(n+3)}{2}$. Call this two marks.

$\mu = E(X) = \int_0^1 xf(x)\,dx = C\int_0^1 x^3(1-x)^n\,dx$.

$$\int_0^1 x^3(1-x)^n\,dx = \frac{3!n!}{(n+4)!} = \frac{6}{(n+1)(n+2)(n+3)(n+4)}.$$

$$\mu = \frac{(n+1)(n+2)(n+3)}{2} \cdot \frac{6}{(n+1)(n+2)(n+3)(n+4)} = \frac{3}{n+4}.$$

For mode, differentiates $f(x)$ $f'(x) = C[2x(1-x)^n - nx^2(1-x)^{n-1}]$.

Sets equal to 0, $x(1-x)^{n-1}[2(1-x) - nx] = 0 \implies 2 - 2x - nx = 0 \implies M = \frac{2}{n+2}$.

Part (ii) [3]

Considers the difference $M - \mu = \dfrac{2}{n+2} - \dfrac{3}{n+4}$.

$$M - \mu = \frac{2(n+4) - 3(n+2)}{(n+2)(n+4)} = \frac{2n + 8 - 3n - 6}{(n+2)(n+4)} = \frac{2-n}{(n+2)(n+4)}.$$

Given $n > 2$, then $2 - n < 0$, which implies $M - \mu < 0$. Concludes $\mu > M$.

Part (iii) [8]

The median m satisfies $\displaystyle\int_0^m f(x)\,dx = \frac{1}{2}$.

To show $M < m < \mu$, it is sufficient to show $\displaystyle\int_0^\mu f(x)\,dx > \frac{1}{2}$ and use the previous result.

$$\int_0^\mu x^2(1-x)^n\,dx = \left[-\frac{x^2(1-x)^{n+1}}{n+1} - \frac{2x(1-x)^{n+2}}{(n+1)(n+2)} - \frac{2(1-x)^{n+3}}{(n+1)(n+2)(n+3)}\right]_0^\mu.$$

Substituting the constant C and $\mu = \frac{3}{n+4}$ yields:

$$F(\mu) = 1 - \left(1 - \frac{3}{n+4}\right)^{n+1}\left[1 + \frac{3(n+1)}{n+4} + \frac{9(n+1)(n+2)}{2(n+4)^2}\right].$$

For $n = 3$, $F(\mu) = 1 - (\frac{4}{7})^4[1 + \frac{12}{7} + \frac{180}{98}]$. This terrible thing evaluates to $1 - \frac{223 \cdot 256}{7^6} = 1 - \frac{57088}{7^6}$.

$7^6 = 117649 > 2 \cdot 57088$, so:

Since $F(\mu) > 0.5$, it follows that $m < \mu$.

Part (iv) [1]

For $x \in (0, 1]$, $x, 1 - x < 1$. So $f(x) = a^n$ for $a \in (0, 1)$. So $f(x) \to 0$ as $n \to \infty$ for all x except $x = 0$. So the PDF becomes a spike at $x = 0$ and hence all measures of central tendency approach 0. Any valid explanation. Sketch is also acceptable.

Question 10

Part (i) [6]

States that $X_1 > k$ means the first k balls drawn are all blue.

Counts the total ways to place the a red balls among the n positions: $\binom{n}{a}$.

For $X_1 > k$, all a red balls must lie in the last $n - k$ positions.

Counts these arrangements: $\binom{n-k}{a}$.

Hence $P(X_1 > k) = \frac{\binom{n-k}{a}}{\binom{n}{a}}$ for $0 \le k \le b$.

Uses $E(X_1) = \sum_k P(X_1 > k)$ and the identity to obtain $E(X_1) = \frac{\binom{n+1}{a+1}}{\binom{n}{a}} = \frac{n+1}{a+1}$.

A more standard probability argument would also work, leading to a bunch of factorials and then rearranging them algebraically. This is fine.

Part (ii) [6]

States that $X_r > k$ means at most $r - 1$ red balls occur among the first k draws.

Counts arrangements of the a red balls among the n positions.

For $X_r > k$, exactly j red balls (with $0 \le j \le r - 1$) appear in the first k positions.

Counts these arrangements as $\binom{k}{j}\binom{n-k}{a-j}$.

Hence $P(X_r > k) = \frac{\sum_{j=0}^{r-1}\binom{k}{j}\binom{n-k}{a-j}}{\binom{n}{a}}$.

Substitutes into $E(X_r) = \sum_{k\geq 0} P(X_r > k)$ and evaluates to obtain $E(X_r) = \frac{r(n+1)}{a+1}$.

Part (iii) [4]

Sets $a = 2$, so there are exactly two red balls.

Let their positions be $X_1 = i$ and $X_2 = j$ with $1 \leq i < j \leq n$.

Condition $X_2 - X_1 = k$ implies $j = i + k$, giving $n - k$ possible pairs.

Since the total number of placements is $\binom{n}{2}$, concludes $P(X_2 - X_1 = k) = \frac{n-k}{\binom{n}{2}} = \frac{2(n-k)}{n(n-1)}$ for $1 \leq k \leq n - 1$.

Part (iv) [4]

States that with replacement the probability of drawing a red ball on each draw is $p = \frac{a}{n}$.

Notes that the expected number of draws needed for one red ball is $1/p = \frac{n}{a}$.

Uses linearity of expectation to obtain $E(Z_r) = r\frac{n}{a} = \frac{rn}{a}$. They can also jump to this immediately by recognising a negative binomial but that's technically not on the spec.

Compares with $E(X_r) = \frac{r(n+1)}{a+1}$ and shows that if $p = \frac{a}{n}$ remains constant then both expressions tend to $\frac{r}{p}$ as $n \to \infty$, so $E(Z_r) - E(X_r) \to 0$.